

Smart Home Builder · Premium Course

3 live sessions · 2 hrs each · 10-14 yrs · from KES 9,000

Premium course outline **PREMIUM**

Overview

Kids build a working smart home model with three integrated systems, wired into one house.

DETAIL

Track	Arduino Robotics & Electronics
Format	3 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	10-14 yrs
Level	Intermediate → Advanced
Price	from KES 9,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 1,000: Small 5 V fan, Servo door-lock bracket, Cardboard house template
Standalone	No
Prerequisites	Traffic light simulation OR Button controlled LED (free)

Programme

1. **Smart Lighting.** Photoresistor + relay auto-lights rooms at dusk; a PIR motion sensor activates hallway lights.
2. **Climate Control.** DHT11 temperature sensor triggers a fan via relay; an LCD displays the current temperature.

3. **Security System.** Ultrasonic door sensor + buzzer alarm + servo-powered door lock, all wired into one house model.

What they take home

 Cardboard smart house model with all circuits installed

Why parents love it

“Your child builds a miniature version of the smart home tech you see advertised — and actually understands how it works.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Smart Home Builder**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.